西南交通大学-利兹学院本科毕业设计(论文)开题报告

毕业设计论文题目	A Study on a Large-Model-Based Intelligent Question-Answering System for the Overhead Contact System Field			
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专　业	Electronic and Electrical Engineering			
校内毕业设计论文	校内导师姓　名	黄德青	导师职称	
选题类型	☐ 毕业论文(理论研究)　☐ 毕业论文(应用研究) ☐ 毕业设计(工程设计)　☐ 毕业设计(软件设计)			
题目来源	☐ 教师自拟　□ 生产实践 ☐ 科研课题　□ 学生自拟　□ 其他			
资料情况	毕业设计指导日志　□ 填写　□ 未填写			

选题意义和国内外研究现状分析

Significance

Chinese railway overhead contact system standards contain specialised terminology, numerical limits and edition-dependent conditions. Keyword searches can miss paraphrased questions, while fluent model answers can omit qualifications or cite an inappropriate edition. A document-grounded question-answering system should therefore expose the evidence and its location, preserve standard identifiers and years, and explain when the selected corpus cannot support an answer. The intended benefit is more traceable study and document consultation; the system will not certify compliance or authorise railway operations.

International research

Retrieval-augmented generation connects language generation with external evidence [1]. BM25 provides a lexical baseline [2], whereas dense passage retrieval uses vector representations for semantic matching [3]. Their different matching mechanisms justify an empirical comparison with score-based Hybrid retrieval rather than assuming that either method is best for technical standards. Query-rewriting research highlights the importance of forming a useful retrieval query [4], motivating limited rule-based follow-ups in this project. Irrelevant-context research [5] motivates evidence-relevance checks, although its arithmetic findings do not establish railway performance. RAGAs distinguishes retrieval quality, faithfulness and answer quality [6]; this project will assess semantic support manually rather than introduce an automatic model-based judge.

Chinese-language research and project focus

Feng et al.'s Chinese-language survey reviews retrieval timing, retrieval methods, evidence use and evaluation [7]. It supports examining retrieval and generation together, but does not establish effectiveness for the standards selected here. The proposed study will investigate three practical questions: whether Hybrid retrieval improves evidence recall, whether edition filtering reduces wrong-edition answers, and whether an Evidence Gate reduces unsupported answers without excessive refusal. The contribution will be a curated corpus, a reproducible engineering comparison and an analysis of failures and limitations; no new model-training method is claimed.

主要研究目标和研究内容	**Aim and research questions** The aim is to design and evaluate a local RAG system for Chinese overhead contact system standards. RQ1 compares retrieval methods, RQ2 evaluates edition filtering, and RQ3 evaluates evidence gating and refusal. **Objectives and system design** O1: By Week 11, curate 20–30 usable documents, recording provenance, permissions, hashes, identifiers, editions and extraction quality. Seek two families with two formal editions each; distinguish duplicates from editions and exclude unsuitable scans/drafts. O2: By Week 16, verify PDF/DOCX parsing, chunking, SQLite, retrieval, DeepSeek, rule-based follow-ups and Streamlit integration. Locate evidence by PDF file-page or DOCX paragraph/table. Refuse missing editions and clarify unspecified editions where multiple candidates exist. O3: By Week 17, freeze 100 source-checked questions: 20 development items (16 answerable) and 80 test items (64 answerable). Record required facts, units, conditions and evidence spans; keep related facts and dialogues within one split. O4: By Week 20, compare B1 BM25, B2 Dense, B3 Hybrid, A1 Hybrid without edition checks, and A2 Hybrid without the retrieval gate. A1 retains standard-family filtering; A2 retains prompts, citation validation and error handling. All five use the same frozen corpus, questions and generation configuration. O5: By Week 24, deliver the system, reproducibility records, report, Portfolio and video; attend the Week 18 Symposium and prepare for the Weeks 25–27 viva. **Method and evaluation** Use consistent Chinese tokenisation and a small domain dictionary, preserving identifiers, technical terms, numbers and units. For Hybrid retrieval, min–max normalise BM25 and cosine scores over eligible chunks and combine them as (1−w)B+wD. Constant positive scores map to one; other constants map to zero. Select w from {0.25, 0.50, 0.75} using development Recall@5, then MRR@5, with ties favouring 0.50. Select gate thresholds on development data only; freeze all settings before testing. Lexical coverage is the proportion of distinct query content tokens matched in passage tokens, excluding identifiers and fixed stopwords; empty sets score zero. The gate accepts sufficient coverage or raw cosine similarity. Metadata, citation numbers and literal excerpts are checked automatically; semantic support requires human judgement. Recall@5 and MRR@5 use pre-gate rankings for all 64 answerable test items. Correctness is fully correct answers/64; false refusal is refusals or unnecessary clarifications/64. Refusal recall is justified non-answering/16 unanswerable items. Citation support is supported cited claims/all cited claims; report citation coverage separately. Wrong-edition rate uses all unambiguously edition-specified items, including missing editions; report availability strata and answer coverage. API/format failures are errors, not justified refusals; zero denominators are N/A. Unsupported-answer incidence uses all 80 test items. Report counts, paired differences, uncertainty, latency and failures. Targets, not results: Recall@5 and correctness ≥0.80, citation support ≥0.90, refusal recall ≥0.80 and false refusal ≤0.10. The main experiment contains 80×5=400 executions. Manual scoring will use a shared rubric and masked configuration labels. **Feasibility and boundaries** An existing BM25 prototype and synthetic DeepSeek checks support initial feasibility; embedding access and formal evaluation remain unverified. Only permitted text may leave the device. Insufficient edition pairs require a limited case study, retaining A1. Reuse CPU hardware and APIs; exclude training, OCR, drawings, agents, knowledge graphs, hardware development and deployment. Acknowledge reused code and AI assistance.

工作计划及进度安排

Research weeks follow the signed ELEC3801 Proposal. Later semester dates and Chinese-side assessment dates follow official notices; these numbered weeks are not a newly approved institutional calendar. The dependency chain is corpus curation → benchmark authoring → configuration freeze → evaluation → report/submission, with integration also required before freezing.

Weeks	Tasks and planned outputs
5–8	T01–T04: scope, initial literature, sample/permission audit and Proposal review. T02 literature work continues to Week 10.
9–10	T05/T06: verify environment, baseline and separate embedding access; begin permitted corpus curation and index checks.
11	T06 / M2: audit the 20–30-document target, usable edition pairs, source locations and BM25 baseline; record unmet criteria explicitly.
12–15	T07/T08/T09: author and source-check questions while verifying edition handling and evidence checks. Study Dense/Hybrid behaviour using development cases.
16	T07/T10 / M3: complete the reviewed question set and end-to-end integration/regression checks, including follow-ups and API failures.
17	T11/T18 / M4: development-only parameter selection; freeze corpus, labels, code, prompts, models and five configurations; prepare Symposium material.
18	T18/T12 / MS: attend the compulsory Symposium and record feedback; begin controlled evaluation after the freeze is ready.
19–20	T12 / M5: complete five comparisons, retrieval/edition metrics, manual answer/citation/refusal scoring and failure analysis. Start T13 after results are checked.
21	T13/T14 / M6: complete the report draft, learning-outcome evidence and demonstration video; prepare Portfolio material before the break.
22–23	Planned project break, 1–14 February 2027. No technical milestones are scheduled; course journal/supervision obligations remain subject to course requirements.
24	T15/T16 / M7: final reproducibility and submission checks; complete report, Portfolio, video access and Chinese-side materials required at the notified checkpoints.
25–27	T17: prepare the presentation and defend the methods, results, limitations and individual contribution.

T16 maintains genuine weekly journals, supervision records and ten Review Documents throughout the applicable working weeks. Planned effort is 372 hours through Week 24 plus 28 hours for the viva, totalling 400 hours; these are budgets, not completed hours. Preserve 36 hours for annotation, 40 for evaluation and 48 for journals/reviews/supervision.

One development pass adds 100 executions to the 400 main test executions: at most 500 first-attempt generations, with a further 100-call reserve. Early refusals may avoid generation; log actual calls, failures and retries separately. Embedding costs are separate. The £50 plan allocates £35 to generation, £10 to embeddings and £5 contingency. Reassess the budget at £40, preserve difficult/negative cases and source checking, and avoid routine repeated test sweeps.

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[1] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.
[2] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333–389, 2009. doi: 10.1561/1500000019.
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<tr><td colspan="2">指导教师意见

指导教师签字:
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<tr><td colspan="2">专家组评审意见

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